INSTITUTE ERP APPS

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Smart Institute CRM & ERP System

A comprehensive operational hub with Role-Based Access Control, Academic Administration, Attendance, and Billing Management.

VERSION

v2.0 (Fully Developed State)

AUTHOR

Development Team

DATE CREATED

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STATUS

Implemented & Validated

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1. Document Overview

This Business Requirements Document (BRD) formalizes the design scope, features, and technical structures implemented within the **Smart Institute CRM & ERP System**. The project is an enterprise-grade ERP application designed for educational institutes, tutorials, coaching centers, and schools to streamline daily operations.

Unlike standard administrative tools, this system implements a unified relational schema to connect academic trackers, financial records, student profiles, and multi-role operations. By digitizing key business procedures, it removes operational silos and improves administrative speed.

Project Purpose

To establish a multi-branch web application that connects 8 distinct roles with customized dashboards, providing real-time data flow for admissions, class attendance, fees collection, notices, and library books.

> **Note on Implementation State:** The current project is a completed and functional React-Node.js application. All features described in Section 3 are fully compiled, seeded with realistic datasets, and deployed securely.

2. Business Objectives & Stakeholder Map

The core business requirements focused on resolving three main pain points: administrative delay, lack of accountability in role duties, and data sync lag between courses, fee ledgers, and attendance charts.

Key Business Metrics

- **Administrative Overhead:** Targeted to reduce manual record entry and document compiling by 60% through digital visitor, registration, and attendance entries.
- **Financial Tracking Accuracy:** Enable 100% visibility of due payments, collection logs, and receipt generations directly within Accountant dashboards.
- **Academic Transparency:** Real-time syllabus and course progress tracking accessible by student, teacher, and admin.

Stakeholder Roles & Permissions (RBAC)

The CRM implements a strict Role-Based Access Control (RBAC) mechanism. Below is the mapped distribution of roles and system actions:

USER ROLE	ASSIGNED OPERATIONAL PERMISSIONS & FOCUS
Super Admin	Full data read/write across all branches, system-wide analytics, audit logs viewing, user creation, and configuration settings.
Branch Admin	Manages local branch courses, allocates batches, assigns faculty, creates exam timetables, posts notices, and generates academic reports.
Faculty	Views assigned batches, marks daily attendance, logs syllabus progress, assigns homework/assignments, and inputs exam grades.
Accountant	Creates fee structures, logs invoice payments, issues receipts, tracks outstanding dues, and generates financial summaries.
Receptionist	Captures public and walk-in enquiries, logs physical visitor check-ins, tracks follow-up status, and initiates registrations.
Librarian	Manages library book inventory, conducts books directory searches, and records issues, returns, and overdue fine logs.

USER ROLE	ASSIGNED OPERATIONAL PERMISSIONS & FOCUS
Student	Learner workspace to view schedules, check attendance stats, track syllabus progress, submit assignments, pay fees, and view marks.
Parent	Linked dashboard providing overview of student's academic standing, fee dues, attendance records, and notices.

3. Implemented Modules & Scope Details

This section outlines the detailed functional requirements that have been successfully built, validated, and integrated within the application codebase:

3.1 Student Registration & Enquiries

- **Public & Admin Forms:** Standardized public enquiry system (`PublicEnquiry.jsx`) allowing visitors to submit admission interest. Receptionists review, follow up, and promote enquiries to official Registrations (`Registrations.jsx`).
- **Credential Auto-Generation:** Successfully auto-generates secure user log-in credentials upon registration approval, categorizing them into Student or Parent tables.

3.2 Course & Batch Management

- **Structured Curriculum:** Setup of standard Courses (Board, Class Range, Targets) and allocating multiple Batches with defined capacities, schedules, and active timings.
- **Faculty Assigns:** Assigning multiple teachers to batches dynamically.

3.3 Syllabus & Progress Tracker (Core Feature)

- **Course Syllabus Schema:** Chapters are broken down into modules and specific topics. Supports syllabus creation, edits, and archiving.
- **Daily Class Tracker:** Faculty Daily Tracker page (`FacultyDailyTracker.jsx`) lets teachers log daily lessons and mark topics as completed, updating batch progress in the database.
- **Student Syllabus Watch:** Students can view their exact syllabus completion percentage, showing transparent academic velocity.

3.4 Attendance Management System

- **Daily Grid markings:** Custom grid interfaces allowing teachers to record daily batch attendance. Supports marking students as present, absent, late, or excused.
- **Reports & Metrics:** Monthly calculation of attendance rates with active dashboards warning students/parents of low attendance scores.

3.5 Exam & Results Module

- **Test Configuration:** Defines Unit tests, Mid Terms, or Final Exams per batch.
- **Grades Entry:** Fast grades grid for faculty to insert student results, calculating ranks, percentages, and passing results instantly.

3.6 Fee Structure & Ledger Tracking

- **Fee Creation:** Establish tailored structures for courses, allocating terms (e.g. Admission Fee, Term 1, Term 2).
- **Accountant Ledger:** Invoices generated automatically. Accountants record manual payments, generate official invoices, and print receipts.
- **Student Billing Portal:** Complete Student Fees dashboard (`StudentFees.jsx`) showing clear balance details, paid ledgers, and simulating payment routes.

3.7 Library Directory & Issue System

- **Book Directory:** Inventory search index showing book title, author, copies, and location.
- **Issue & Return Logs:** Librarians track books checked out by students and teachers, marking due dates, returns, and assessing overdue fines.

3.8 Reception Center & Visitor Logs

- **Visitor Logs:** Receptionists track check-in and check-out times of visitors, logging physical purposes.
- **Call Log & Tasks:** Tracks incoming/outgoing communications, follow-up states, and action plans.

3.9 Homework & Assignments Portal

- **Assignment Creation:** Faculty upload assignment titles, descriptions, and materials (links or attachments) tied to specific batches.
- **Submissions Tracker:** Students view due assignments and upload work files. Submissions database logs timestamped entries.

3.10 Notices & Notice Board

- **Announcements:** Admins and teachers publish formal notices categorized by type (Exams, Events, Holiday, Fees).
- **RBAC Targeting:** Visibility filters targeting specific roles (All, Students, Faculty, Parents) to avoid information clutter.

3.11 System Settings & Security Audits

- **App settings:** Super Admins customize company titles, contact details, standard boards, and branch profiles dynamically.
- **Security logs:** Keeps strict database records of administrative operations (logins, database updates) for system integrity.

4. Technical Stack & Database Architecture

The application is designed using a robust three-tier architecture with stateful relational data integrity. It is optimized for zero-downtime hosting on cloud providers such as Render and Supabase.

Frontend Tier

React.js (v18), Vite, Vanilla CSS, Tailwind, React Router DOM (v6), Context API

Backend Tier

Node.js Express Framework, Sequelize ORM, JWT Authentication, Multer

Database Tier

PostgreSQL (Supabase Hosting) / MySQL compatible schema, Sequelize Migrate

Relational Schema & Key Model Associations

The backend database implements structured relational tables. Key associations compiled in the code include:

PRIMARY MODEL	RELATIONSHIP TYPE	CONNECTED MODEL & KEY USAGE
User	HasMany (1:N)	Enrollments (student_id), FeePayments (student_id), IssuedBooks (student_id), Salaries (user_id), Leaves (user_id), Complaints (user_id), Certificates (student_id)
Course	HasMany (1:N)	Enrollments (course_id), BatchProgress (course_id)
Batch	BelongsToMany (M:N)	Faculty (through BatchFaculty) to map class assignments. HasMany BatchProgress entries.
BatchProgress	HasMany (1:N)	Attendance records (batch_progress_id) to link markings directly to specific class lessons.
Branch	HasMany (1:N)	Users (branch_id) to enable dynamic multi-branch ERP structures.
LibraryBook	HasMany (1:N)	IssuedBooks (book_id) to manage library inventory checks.

Data Security & Authorization Flow

All client requests transit through a middleware layer verifying JSON Web Tokens (JWT). Roles are checked at the API level (Express routers) and dynamically render UI layouts in React pages using the custom context hook `useAuth()`.

5. Scope Verification Matrix: Done vs. Out-of-Scope

The grid below lists the baseline requirements and classifies them into completed deliverables (Done) or items explicitly marked as out of scope for the current phase.

✓ IN SCOPE (FULLY DELIVERED & OPERATIONAL)

Module Group	Implemented Capabilities	Status
Multi-Role Dashboards	8 custom layouts matching RBAC structures.	DONE
Academic Tracker	Courses, batches, lesson markers, syllabus versions.	DONE
Billing & Ledger	Auto invoicing, Accountant collection portal, balance logs.	DONE
Admin Registrations	Physical logs, enquiry reviews, user credentials auto-gen.	DONE
Operations Add-ons	Library catalogs and reception desk management.	DONE

✕ OUT OF SCOPE (EXCLUDED FROM THE WEB SYSTEM FOR NOW)

Excluded Feature	Rationale & Integration Barriers	Classification
Full AI Automation	AI teaching bots, voice calling bots, exam grading engine.	OUT-SCOPE
Hardware Sync	Biometric fingerprint scanners or RFID smart-gates.	OUT-SCOPE
Payroll & Inventory	Staff payroll ledger and school uniforms/book sales tracker.	OUT-SCOPE
Tally/GST Accounting	GST compliance, double-entry banking reconciliation.	OUT-SCOPE
Transport Fleet & Routes	School bus scheduling, route routing stop mappings, vehicle tracking and allocations.	OUT-SCOPE

6. Future Roadmap & Advanced Integrations

To enable modular growth, the current codebase has been designed with clean models and isolated route structures. This prepares the system for the implementation of subsequent operational phases:

Phase 2: Transport & Fleet Management

Development and activation of the transport module, adding the Transport Manager user role and dashboards. This phase will include vehicle tracking schemas, bus driver records, stops allocations, student transport assignments, and live route status updates.

Phase 3: External Automation & n8n

Integration of an automation worker layer (such as n8n.io) to enable automated comparisons of the local syllabus tables against external boards (NEET, JEE, SSC, etc.), auto-archiving legacy chapters, and compiling AI summaries of syllabus updates.

Phase 4: Conversational Messaging & Mobile App

- **WhatsApp & SMS Integration:** Syncing attendance alerts, fee invoice links, and exam result reports to parents via official WhatsApp Business APIs.
- **Native Mobile Applications:** Releasing Android and iOS client wrappers for Students/Teachers to push immediate notifications of assignments, announcements, and results.
- **AI Tutor Assistant:** Implementing local LLM agents to provide study recommendations, auto-correcting test templates, and highlighting student weak topics based on score histories.

7. Assumptions & Constraints

- **Stable Connections:** The deployment model relies on continuous HTTPS APIs between the Render-hosted frontend static asset server and the Render Node.js backend.
- **Data Quality:** Automated syllabus compare actions (Phase 2) require structured JSON/XML API interfaces from board servers. Legacy unstructured boards may require human validation.
- **Browser Dependencies:** All interactive grids (Attendance markers, results input) assume standard HTML5-compatible client browsers.

8. Conclusion

This Business Requirements Document confirms the successful implementation of the **Smart Institute CRM & ERP System**. All core operations—spanning student registrations, 8 RBAC roles, course and batch schedulers, attendance grids, homework, exams grading, billing, receptionist enquiries, and librarian registries—are fully operational and ready for deployment.

> **Sign-off Status:** All core functional milestones are complete. This document is saved as a verified resource for administrative operations.